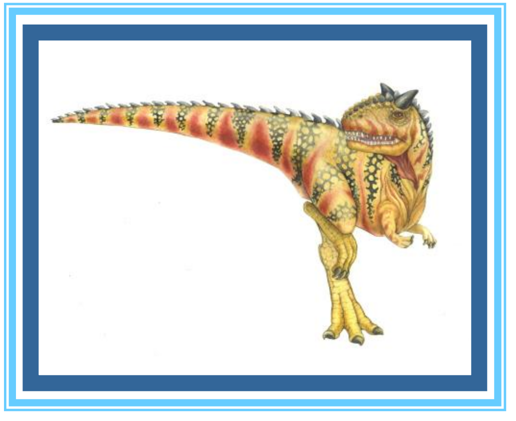
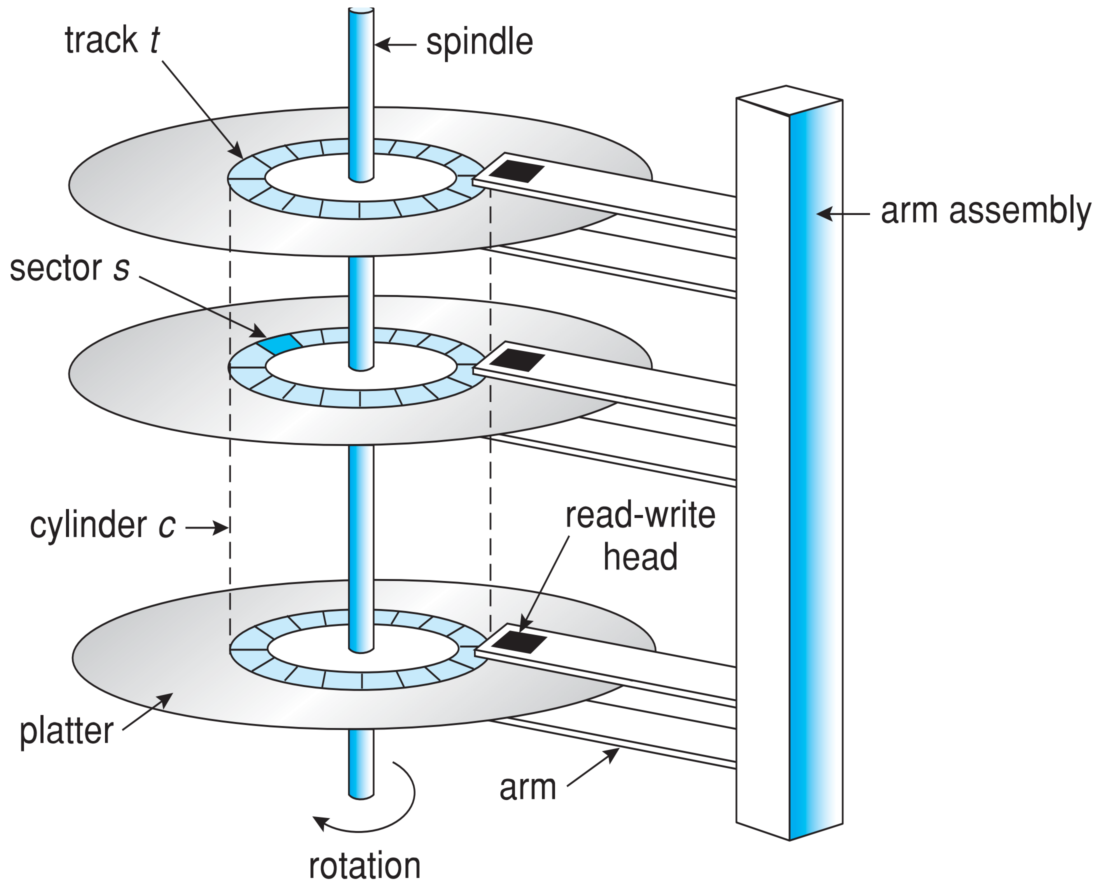
Chapter 11: Mass-Storage Systems



Arhitectura discurilor dure (HDD)

- mediul de stocare format prin “stivuirea” unor *platane* pe un ax care formeaza un *pachet*
- acest pachet este invartit in jurul axului la o viteza de rotatie constanta (5000-10000 rpm)
- blocurile de disc (*sectoare*) situate la aceeasi distanta de centrul unui platan alcatuiesc o *pista*
- setul de piste aflat la aceeasi distanta fata de axul platanelor alcatuieste un *cilindru*
- toate datele dintr-un cilindru pot fi accesate simultan fara miscarea capului de citire
- datele se citesc in multipli de dimensiunea blocului (1KB – 4KB)

Moving-head Disk Mechanism



Citirea blocurilor

- se muta capul de citire deasupra cilindrului care contine blocul de date (*seek*) - cca 5 ms
- se asteapta pana cand discul se roteste a.i. datele sa ajunga sub capul de citite (*rotational delay*) – cca 4ms pt un disc cu 7200 rpm
- se transfera datele (*transfer*) prin alegerea capului de citire de deasupra pistei pe care se afla datele – cca 0.1ms pt 1KB
- timpul mediu de acces random in general

$$t_{seek} + t_{rotatie} + t_{transfer} = 9.1 \text{ ms}$$

- timpul mediu de acces random de pe acelasi cilindru

$$t_{rotatie} + t_{transfer} = 4.1 \text{ ms}$$

- timpul de citire a urmatorului bloc de pe aceeasi pista

$$t_{transfer} = 0.1 \text{ ms}$$

- optimizari posibile: minimizare timpilor de seek si rotatie

HDD Scheduling

- The operating system is responsible for using hardware efficiently — for the disk drives, this means having a fast access time and disk bandwidth
- Minimize seek time
- Seek time $\approx$ seek distance
- Disk **bandwidth** is the total number of bytes transferred, divided by the total time between the first request for service and the completion of the last transfer

Disk Scheduling (Cont.)

- There are many sources of disk I/O request
  - OS
  - System processes
  - Users processes
- I/O request includes input or output mode, disk address, memory address, number of sectors to transfer
- OS maintains queue of requests, per disk or device
- Idle disk can immediately work on I/O request, busy disk means work must queue
  - Optimization algorithms only make sense when a queue exists
- In the past, operating system responsible for queue management, disk drive head scheduling
  - Now, built into the storage devices, controllers
  - Just provide LBAs, handle sorting of requests
    - Some of the algorithms they use described next

Disk Scheduling (Cont.)

- Note that drive controllers have small buffers and can manage a queue of I/O requests (of varying "depth")
- Several algorithms exist to schedule the servicing of disk I/O requests
- The analysis is true for one or many platters
- We illustrate scheduling algorithms with a request queue (0-199)

98, 183, 37, 122, 14, 124, 65, 67

Head pointer 53

Algoritmi de disk scheduling

- incearca sa minimizeze timpul de cautare (*seek time*)
- FCFS – simplu, dar ineficient
  - ex: coada de cereri blocuri aflate pe cilindrii 53, 98, 183, 37, 122, 14, 124, 65, 67 => capetele de citire se vor muta peste 640 de cilindri

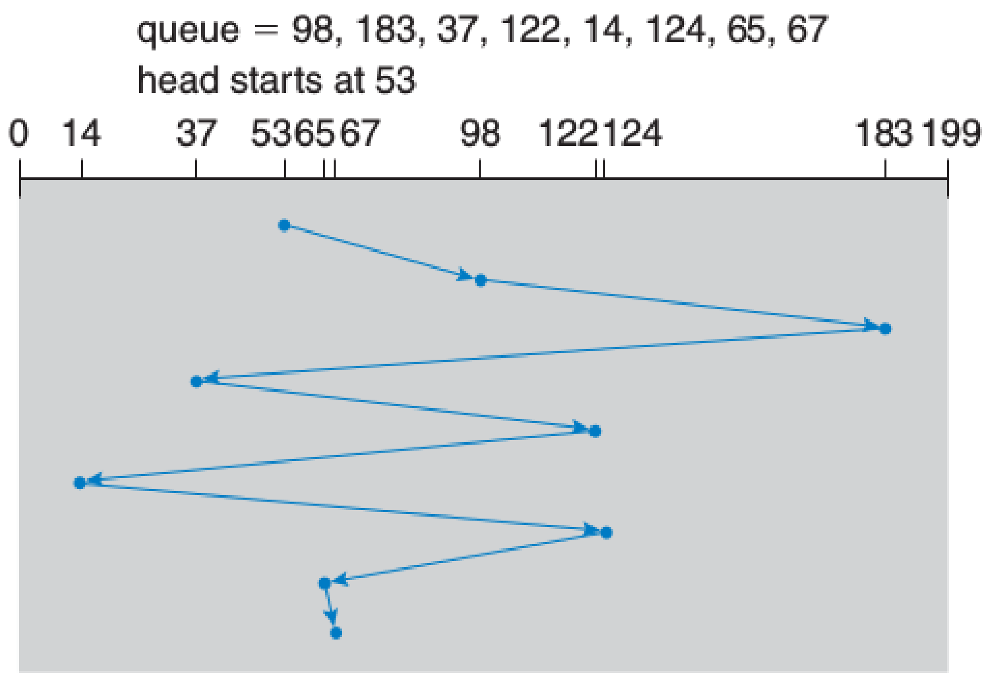


Figure 9.4 FCFS disk scheduling.

SSTF (shortest seek time first)

- serveste cererile de pe cilindrii cei mai apropiati de pozitia curenta a capetelor de citire
- ex anterior: capetele de citire se muta peste 236 de cilindri
- nu e optimal, ex: 53, 37, 14, 65, 67, 98, 122, 124, 183 => 208 cilindri parcursi
- sufera de starvation: daca apar in permanenta cereri in apropierea capetelor de citire, cererile “indepartate” sunt intarziate indefinit

SSTF (shortest seek time first)

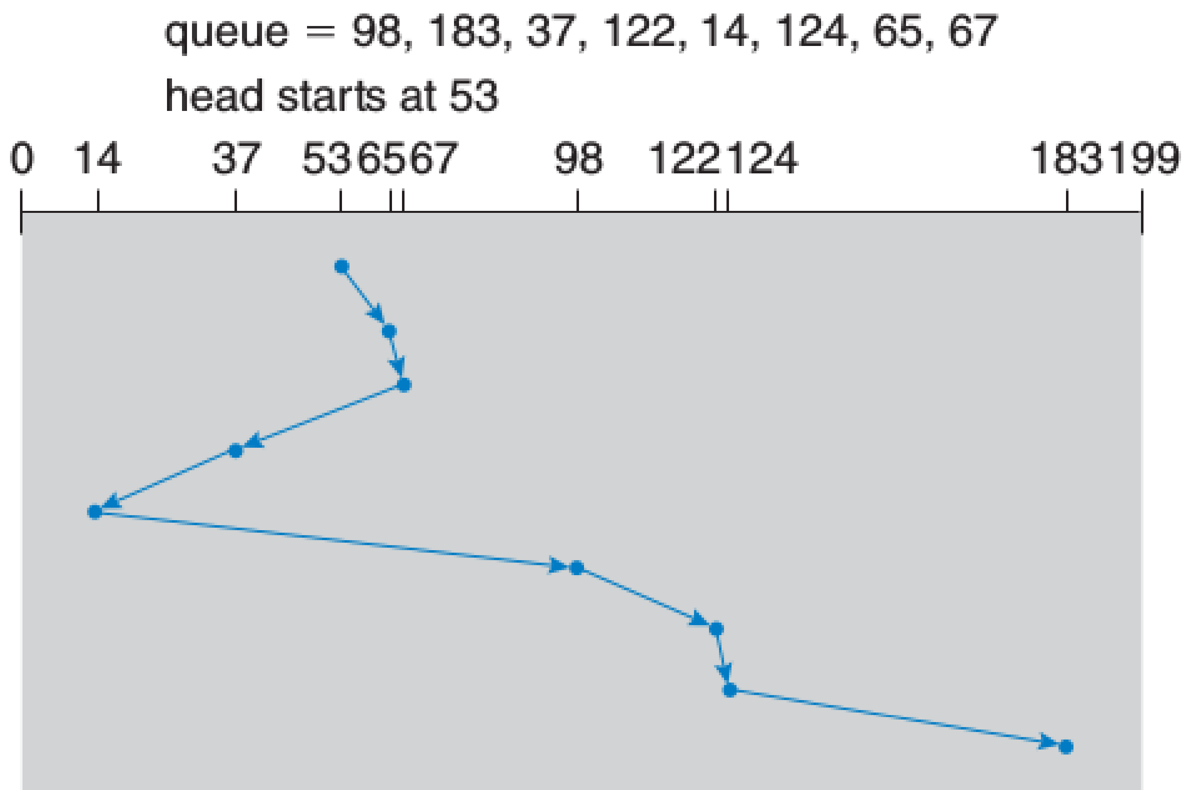


Figure 9.5 SSTF disk scheduling.

Algoritmi de disk scheduling (cont'd)

- SCAN (algoritmul liftului)
  - bratul discului porneste de la un capat al acestuia catre celalalt si serveste cererile intalnite in cale
  - ajuns la capatul discului o ia in sens invers
  - o cerere aparuta chiar inaintea capului de citire e servita imediat
  - o cerere aparuta imediat in spatele capului de citire e intarziata pana se intoace capul de citire in sens contrar

SCAN

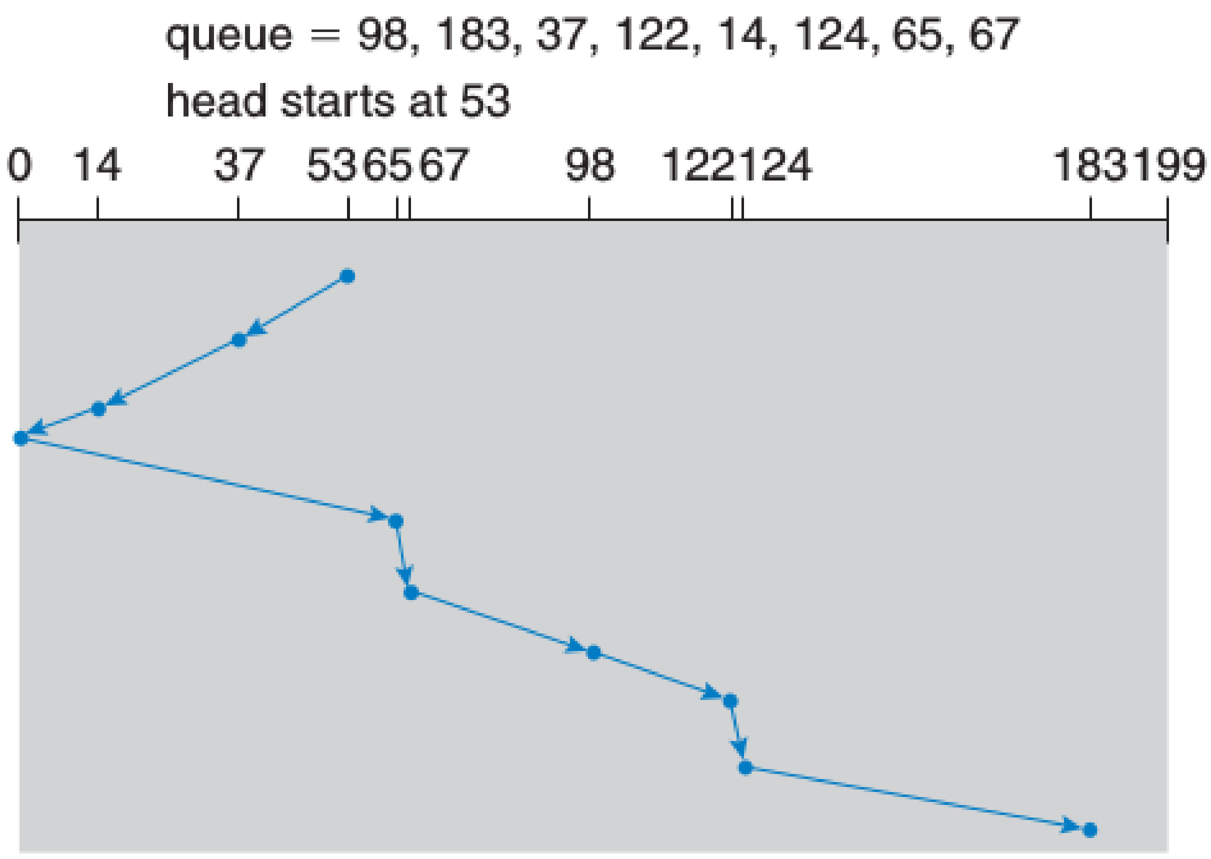


Figure 9.6 SCAN disk scheduling.

Algoritmi de disk scheduling (cont'd)

- C-SCAN (circular SCAN)
  - pt o distributie uniforma a cererilor, cand bratul ajunge la capatul discului si se intoarce, exista relativ putine cereri in fata capului pt ca acestea tocmai au fost tratate
  - densitatea mare de cereri noi e la celalalt capat al discului unde cererile au asteptat cel mai mult
  - ofera timp de asteptare mai uniform (la momentul atingerii capatului discului, capul de citire se intoarce la celalalt capat fara a mai trata cererile din fata capului de citire)
  - trateaza cilindrii discului ca pe o lista circulara
  - ex anterior: bratul se muta peste 183 de cilindri (se considera ca mutarea capului de citire la inceputul discului e o operatie f. rapida)

C-SCAN

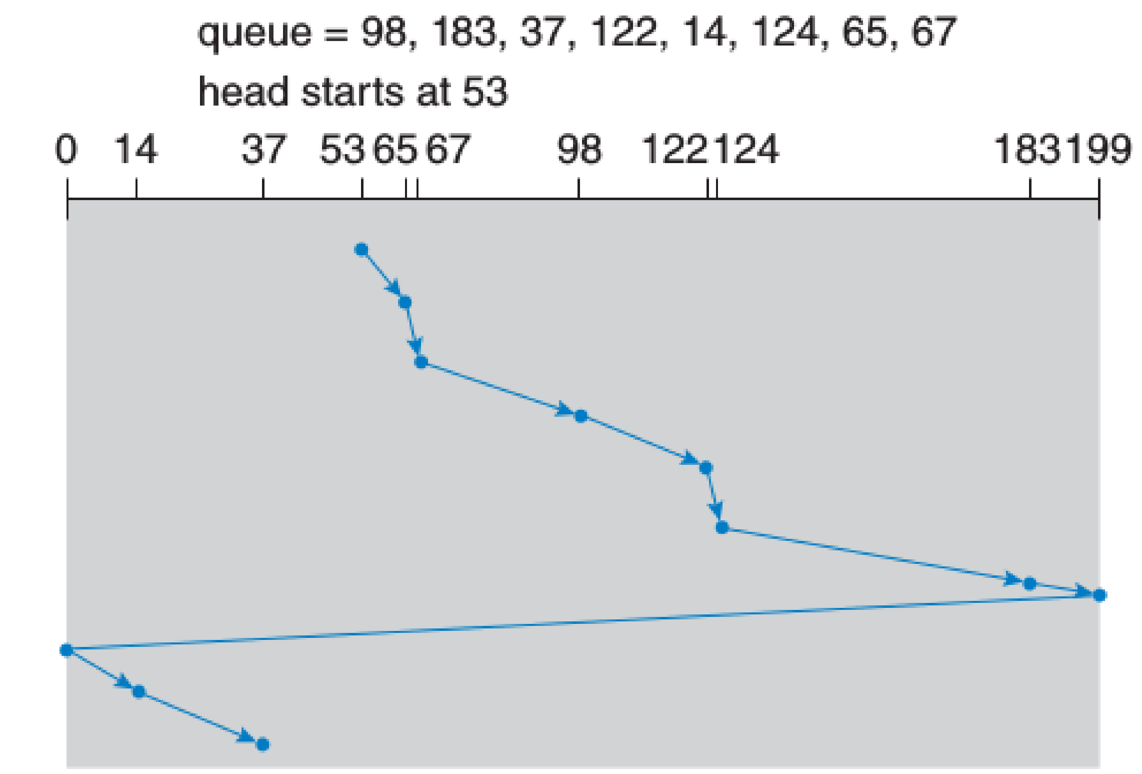


Figure 9.7 C-SCAN disk scheduling.

Algoritmi de disk scheduling (cont'd)

- LOOK
  - SCAN si C-SCAN muta capul de citire peste tot discul
  - o implementare practica ia in calcul doar cererile pt cilindrii situati intre nr minim, respectiv maxim din lista de cereri
  - algoritmii respectivi s.n. LOOK si respectiv C-LOOK
  - ex anterior: la cilindrul 183 bratul se intoarce imediat la cilindrul 14

C-LOOK

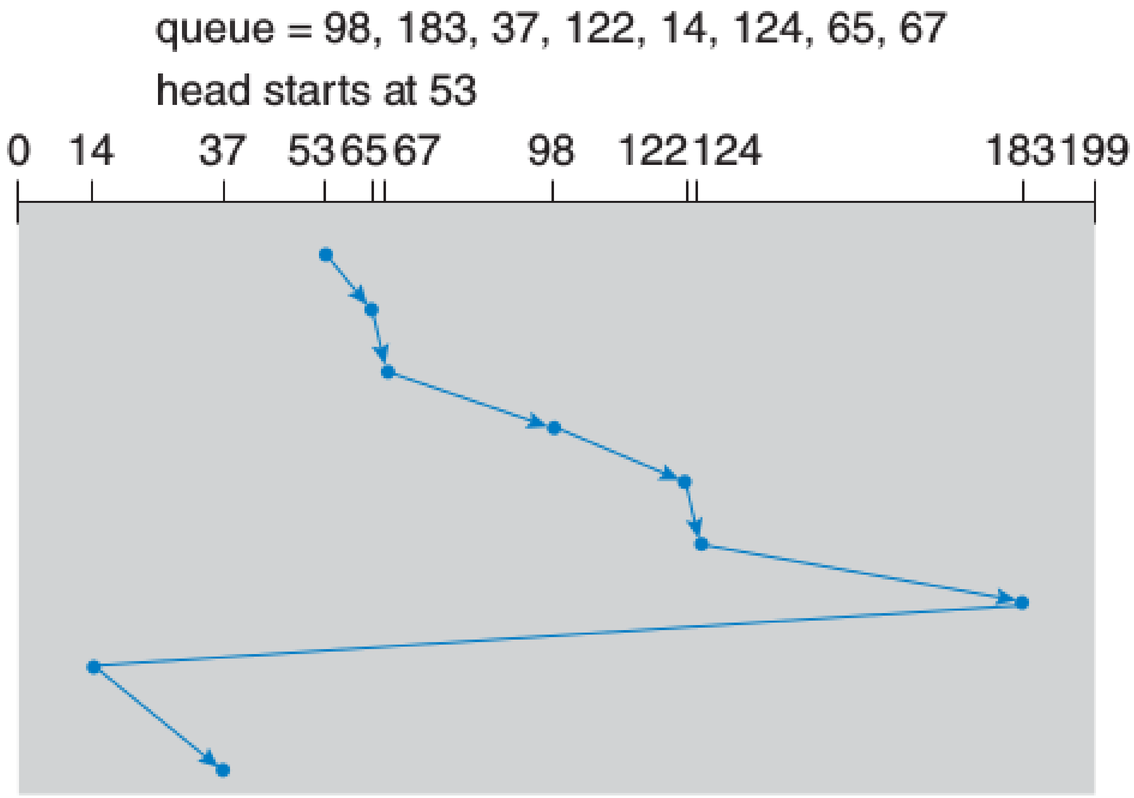


Figure 9.8 C-LOOK disk scheduling.

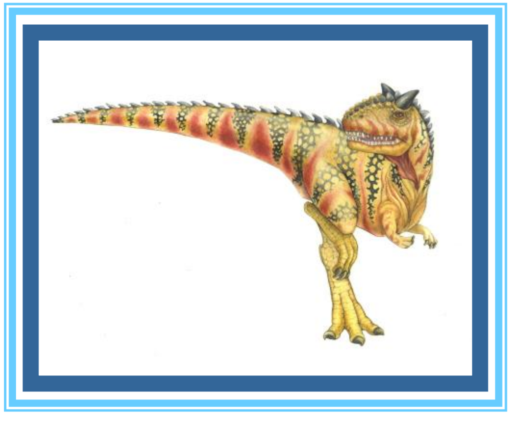
Factori care influenteaza performanta

- nr de cereri si tipul lor
  - daca exista o singura cerere in coada, toti algoritmii se comporta ca FCFS
- metoda de alocare a fisierului
  - alocare contigua – miscare limitata a capetelor de citire
  - alocare indexata – blocurile pot fi imprastiate pe disc => deplasari mari ale capetelor de citire
- plasamentul directoarelor si indexarea blocurilor
  - directoarele sunt accesate frecvent (deschiderea unui fisier pp accesarea unui director)
  - daca intrarea in director e pe cilindrul 1 si datele sunt pe ultimul cilindru, bratul de citire trebuie sa parcurga toti cilindrii
  - daca directorul e plasat pe un cilindru din mijloc, bratul se misca doar jumatare din "latimea" discului in termeni de cilindri
  - caching-ul directoarelor si al indecsilor de blocuri ajuta f. mult la evitarea miscarilor bratului discului

Implementare HW vs SW

- algoritmii anteriori nu iau in calcul timpul rotational care e dificil de optimizat fara cunostinte despre arhitectura HW a discului
- solutie: discurile moderne implementeaza algoritmii anteriori in HW pt a putea optimiza simultan si timpul de cautare si pe cel rotational
- pe de alta parte, sistemul de operare are propriile motive sa implementeze scheduling-ul de disc, de ex:
  - paginarea la cerere e mai importanta decat I/O-ul aplicatiilor
  - scrierile sunt mai importante decat citirile cand cache-urile din sistem nu mai au memorie suficienta
  - pt a preveni pierderea de date, e bine sa se garanteze scrierea unui anumit nr de blocuri de disc inaintea unui eventual crash

Chapter 13: File-System Interface



Fisiere

- abstractie de nivel de sistem de operare pt stocarea persistenta a datelor
- la nivelul cel mai de jos, stocarea persistenta se face pe discuri (mai recent, memorii flash, SSD, NVRAM, etc)
- sistemul de fisiere
  - componenta a sistemului de operare (i.e., parte a kernelului)
  - gestioneaza mediul de stocare persistenta a datelor
  - ofera la nivelul de aplicatie abstractia de fisier si apeluri sistem corespunzatoare
- fisierele
  - concret, containere pt stocarea persistenta a datelor
  - uzual referite prin nume (string ASCII) convertit la o reprezentare interna a kernelului de catre sistemul de fisiere
  - paradigma uzuala de folosire: open – read/write - close

File Concept

- Contiguous logical address space
- Types:
  - Data
    - Numeric
    - Character
    - Binary
  - Program
- Contents defined by file's creator
  - Many types
    - **text file,**
    - **source file,**
    - **executable file**

File Attributes

- **Name** – only information kept in human-readable form
- **Identifier** – unique tag (number) identifies file within file system
- **Type** – needed for systems that support different types
- **Location** – pointer to file location on device
- **Size** – current file size
- **Protection** – controls who can do reading, writing, executing
- **Time, date, and user identification** – data for protection, security, and usage monitoring
- Information about files are kept in the directory structure, which is maintained on the disk
- Many variations, including extended file attributes such as file checksum
- Information kept in the directory structure

File Operations

- **Create**
- **Write –** at **write pointer** location
- **Read –** at **read pointer** location
- **Reposition within file -** **seek**
- **Delete**
- **Truncate**
- ***Open*** **(*F_i*)** – search the directory structure on disk for entry ***F_i***, and move the content of entry to memory
- ***Close*** **(*F_i*)** – move the content of entry ***F_i*** in memory to directory structure on disk

Open Files

- Several pieces of data are needed to manage open files:
  - **Open-file table**: tracks open files
  - File pointer: pointer to last read/write location, per process that has the file open
  - **File-open count**: counter of number of times a file is open – to allow removal of data from open-file table when last processes closes it
  - Disk location of the file: cache of data access information
  - Access rights: per-process access mode information

Open Files (cont.)

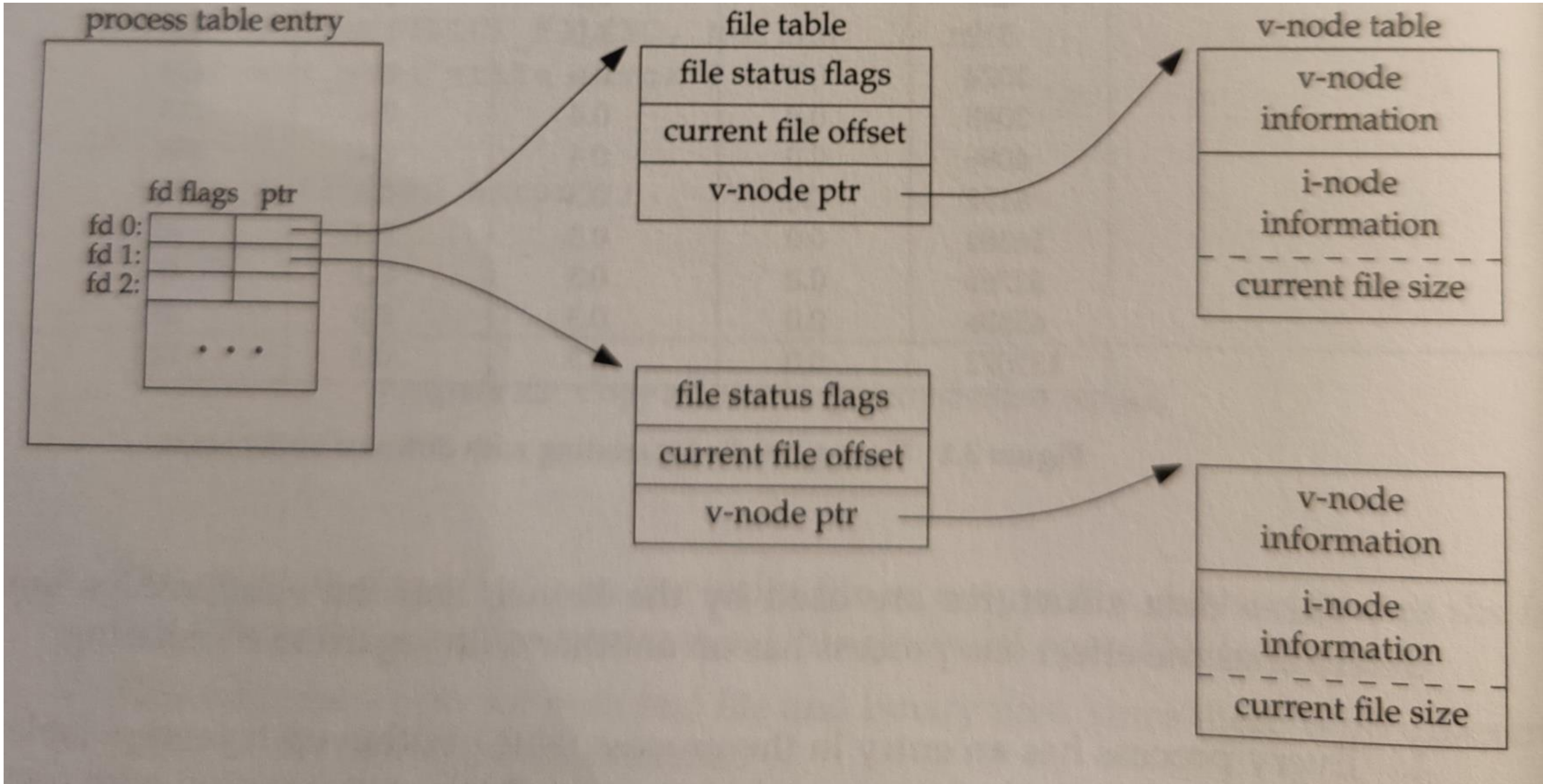


Figure 3.2 Kernel data structures for open files.

File Locking

- Provided by some operating systems and file systems
  - Similar to reader-writer locks
  - **Shared lock** similar to reader lock – several processes can acquire concurrently
  - **Exclusive lock** similar to writer lock
- Mediates access to a file
- Mandatory or advisory:
  - **Mandatory** – access is denied depending on locks held and requested
  - **Advisory** – processes can find status of locks and decide what to do

File Structure

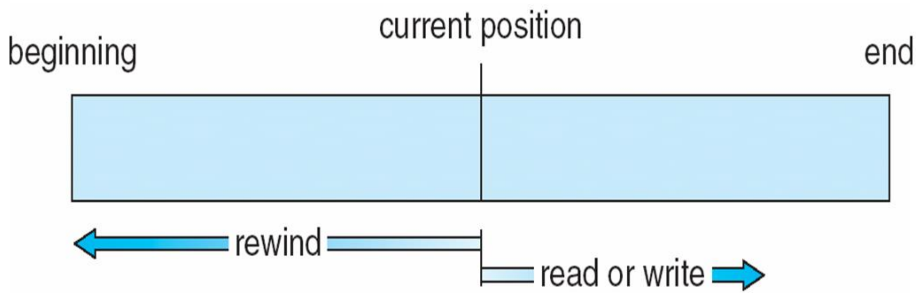
- None - sequence of words, bytes
- Simple record structure
  - Lines
  - Fixed length
  - Variable length
- Complex Structures
  - Formatted document
  - Relocatable load file
- Can simulate last two with first method by inserting appropriate control characters
- Who decides:
  - Operating system
  - Program

Access Methods

- A file is fixed length **logical records**
- **Sequential Access**
- **Direct Access**
- **Other Access Methods**

Sequential Access

- Operations
  - **`read next`**
  - **`write next`**
  - **`Reset`**
  - no read after last write (rewrite)
- Figure



Direct Access

- Operations
  - **`read`** ***n***
  - **`write`** ***n***
  - **`position to`** ***n***
    - **`read next`**
    - **`write next`**
    - **`rewrite`** ***n***

  n = **relative block number**

- Relative block numbers allow OS to decide where file should be placed

Fisiere in Unix

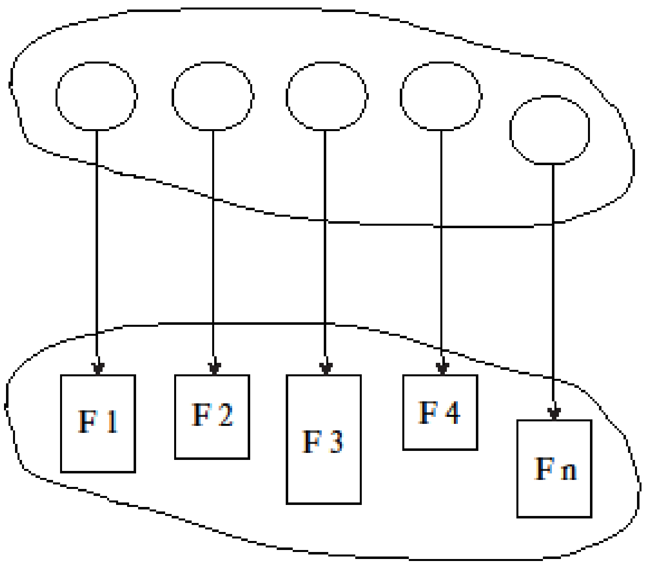
- perspectiva particulara: datele utilizator, organizarea fisierelor in directoare si driverele de echipamente accesibile prin intermediul interfetei sist. de fisiere
- tipuri de fisiere
  - fisiere obisnuite:
    - stocheaza datele utilizator
    - implementate ca un stream/sir de bytes (complet nestructurat)
  - fisiere director (directoare):
    - contin perechi (nume fisier, inode)
    - instituie o structura ierarhica a spatiului de nume pt fisiere
  - fisiere speciale
    - interfata catre driverele de echipamente sau structuri de date speciale ale kernelului (eg, named pipes pt IPC, Unix sockets)
- traditional, toate fisierele se accesau folosind aceleasi apeluri sistem
- ulterior, API special pt lucrul cu directoare (*opendir* – *readdir* – *rewinddir* – *closedir*)

Fisiere Unix la nivel aplicatie

- inainte de accesul la date, un fisier trebuie “deschis” (apel de sistem *open*)
- kernelul aloca un *descriptor de fisier* care este returnat aplicatiei pt folosinta
- intern, kernelul aloca si un obiect corespunzator fisierului deschis care contine un *file pointer* (setat pe 0, imediat dupa *open*) care reflecta offsetul in cadrul stream-ului de bytes
- pozitia file pointer-ului se poate schimba cu *seek*
- *read/write* lucreaza cu byte-ul referit de catre file pointerul current
- doua procese care deschid acelasi fisier folosesc offset-uri diferite in fisier
- descriptorii de fisiere se pot duplica folosind *dup/dup2*, situatie in care file pointer-ul este partajat
- fisierele sunt organizate intr-o ierarhie arborescenta cu o radacina (*root*) unica
  - in general insa, in arborele de fisiere coexista mai multe sisteme de fisiere atasate structurii arborescente de directoare cu ajutorul unei operatii speciale *mount*

Directory Structure

- A collection of nodes containing information about all files



- Both the directory structure and the files reside on disk

Operations Performed on Directory

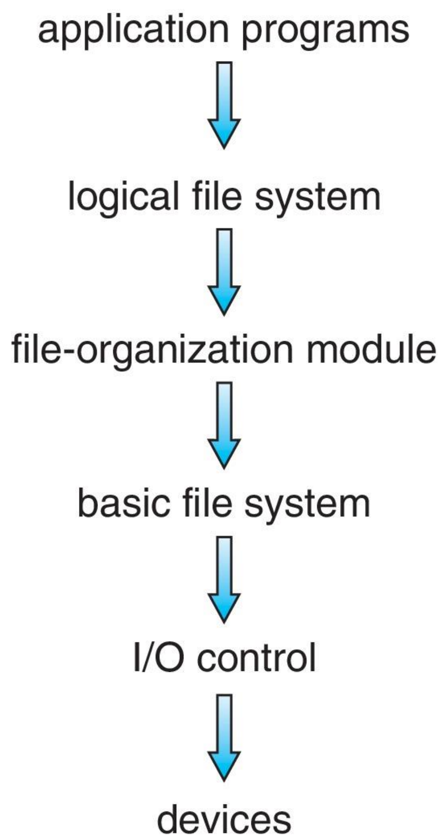
- Search for a file
- Create a file
- Delete a file
- List a directory
- Rename a file
- Traverse the file system

Chapter 14: File System Implementation

File-System Structure

- File structure
  - Logical storage unit
  - Collection of related information
- **File system** resides on secondary storage (disks)
  - Provided user interface to storage, mapping logical to physical
  - Provides efficient and convenient access to disk by allowing data to be stored, located retrieved easily
- Disk provides in-place rewrite and random access
  - I/O transfers performed in **blocks** of **sectors** (usually 512 bytes)
- **File control block** (**FCB**) – storage structure consisting of information about a file
- **Device driver** controls the physical device
- File system organized into layers

Layered File System



File System Layers

- **Device drivers** manage I/O devices at the I/O control layer

  Given commands like

    read drive1, cylinder 72, track 2, sector 10, into memory location 1060

  Outputs low-level hardware specific commands to hardware controller
- **Basic file system** given command like “retrieve block 123” translates to device driver
- Also manages memory buffers and caches (allocation, freeing, replacement)
  - Buffers hold data in transit
  - Caches hold frequently used data
- **File organization module** understands files, logical address, and physical blocks
- Translates logical block # to physical block #
- Manages free space, disk allocation

File System Layers (Cont.)

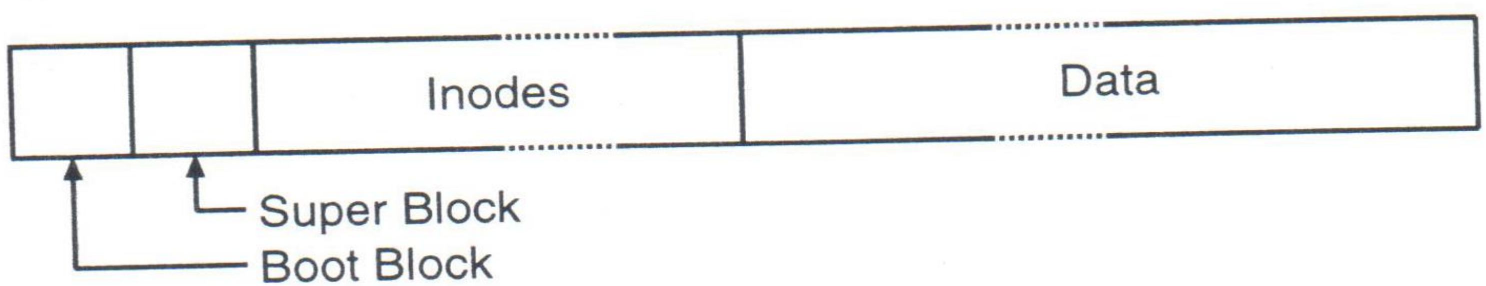
- **Logical file system** manages metadata information
  - Translates file name into file number, file handle, location by maintaining file control blocks (**inodes** in UNIX)
  - Directory management
  - Protection
- Layering useful for reducing complexity and redundancy, but adds overhead and can decrease performance
- Logical layers can be implemented by any coding method according to OS designer

File-System Operations

- We have system calls at the API level, but how do we implement their functions?
  - On-disk and in-memory structures
- **Boot control block** contains info needed by system to boot OS from that volume
  - Needed if volume contains OS, usually first block of volume
- **Volume control block** (**superblock, master file table**) contains volume details
  - Total # of blocks, # of free blocks, block size, free block pointers or array
- Directory structure organizes the files
  - Names and inode numbers, master file table

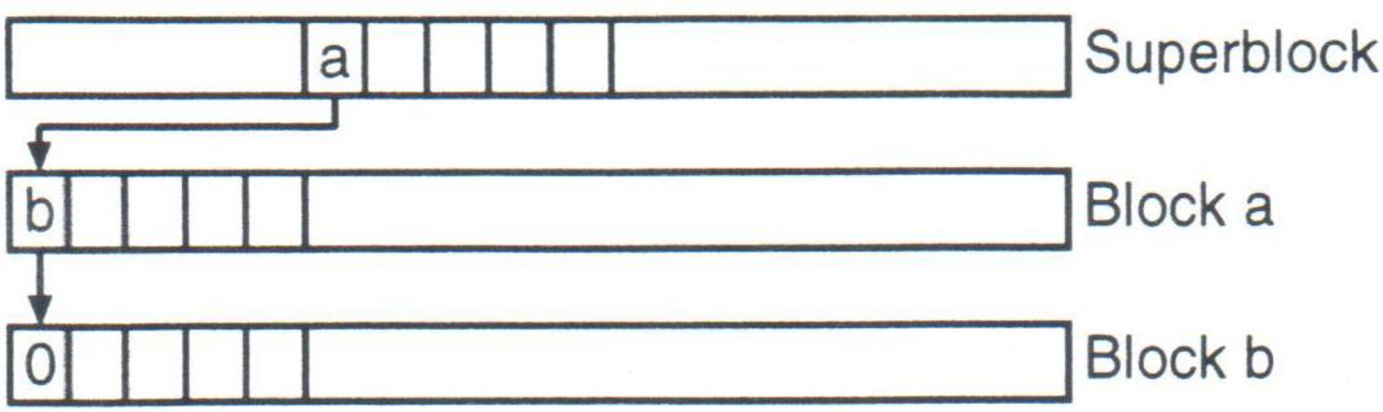
Unix System V

- primul bloc al sistemului de fisiere rezervat codului de boot (chiar daca nu e folosit, rezervarea pastreaza numerotarea restului blocurilor din sistemul de fisiere)
- superblock-ul contine:
  - dimensiunea in blocuri a sistemului de fisiere si a listei de inode-uri
  - nr de blocuri si inode-uri libere
  - lista partiala cu blocuri libere
  - lista partiala cu inode-uri libere
  - in general, cele 2 liste sunt prea mari pt a fi tinute integral in superblock



Unix System V (cont'd)

- uzual, superblock-ul contine vectori care identifica primele inode-uri, respectiv blocuri de date libere
  - cand lista de inode-uri libere e goala, kernelul scaneaza discul pt a gasi inode-uri libere si a reumple lista partiala
  - pt lista de blocuri libere nu se procedeaza similar (pt ca nu se poate decide daca un bloc de date e liber sau nu bazat doar pe continutul sau) => se folosesc liste de blocuri libere inlantuite folosind primul element al listei partiale de blocuri libere din superblock



Inode-uri Unix

- fiecare fisier e reprezentat prin *i-node* (*index node*)
- lista de inode-uri de dupa superblock are lungime fixa si e alocata la crearea sistemului de fisiere (“formatarea discului”)
- inode-ul contine:
  - tipul fisierului
  - drepturile asupra fisierului
  - nr de *link*-uri ale fisierului (i.e., nr de intrari in directoare diferite)
  - dimensiunea fisierului in octeti
  - timpul ultimului acces
  - timpul ultimei modificari a fisierului
  - timpul ultimei schimbari a inode-ului
  - vectorul de adrese de disc

Vectorul de adrese de disc

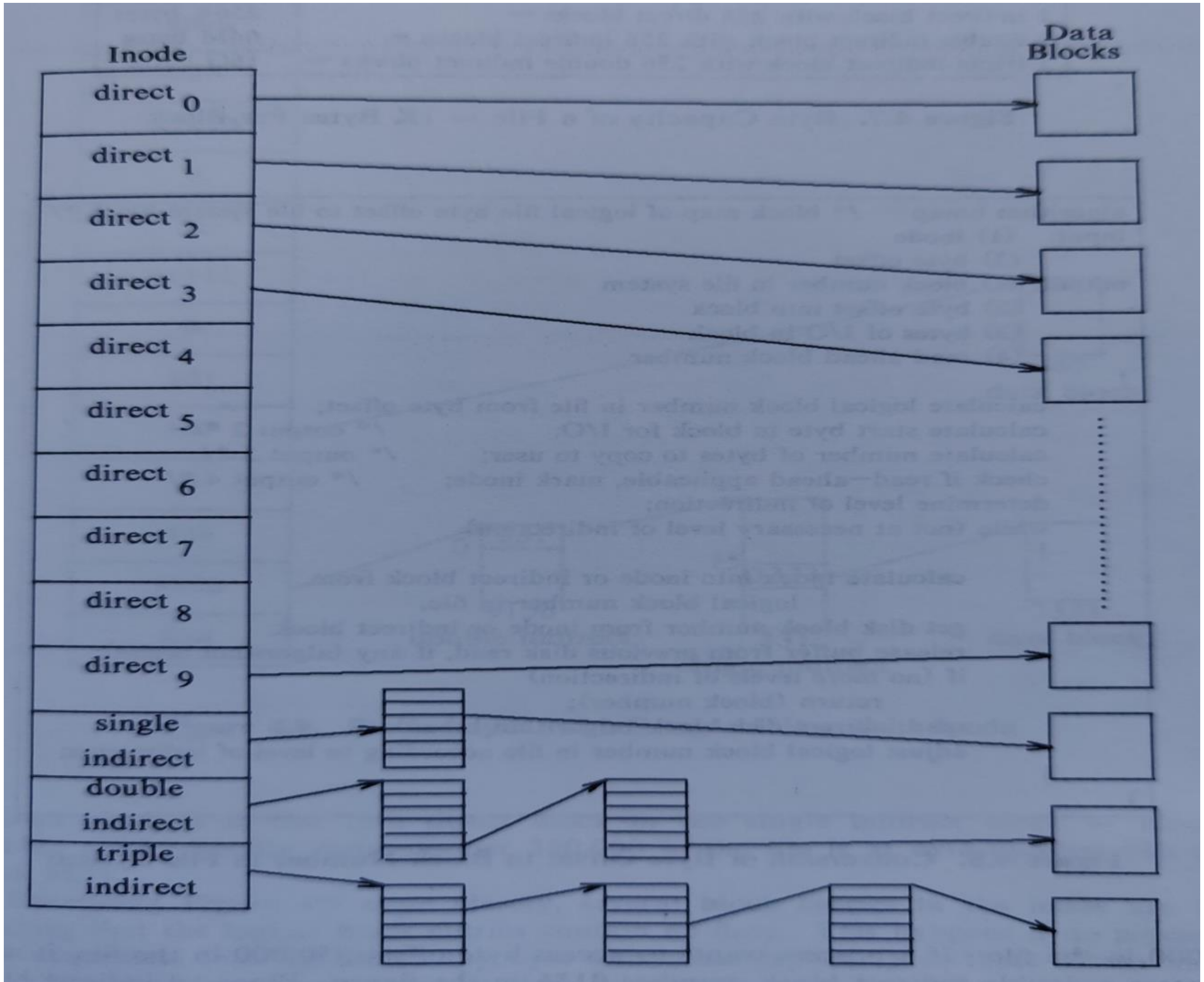


Figure 4.6. Direct and Indirect Blocks in Inode

Exemplu

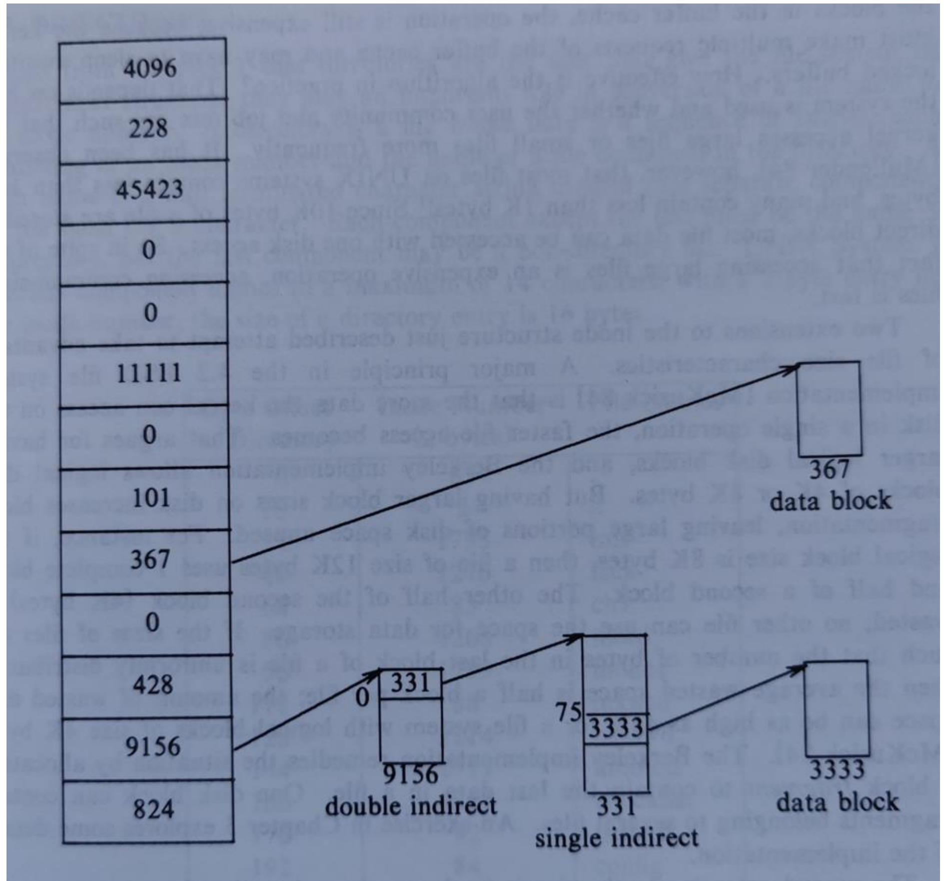


Figure 4.9. Block Layout of a Sample File and its Inode

Calcul dimensiune maxima fisier

- dimensiune bloc disk = 1KB
- adresa bloc disk reprezentata pe 4 octeti

=> 256 adrese indirecte intr-un bloc

- limita impusa de definitia inode 4GB (dimensiunea fisierului reprezentata pe 32 biti)

=> teoretic, fisierele pot avea ceva mai mult de 16GB, practic <= 4GB

10 direct blocks with 1K bytes each =	10K bytes
1 indirect block with 256 direct blocks =	256K bytes
1 double indirect block with 256 indirect blocks =	64M bytes
1 triple indirect block with 256 double indirect blocks =	16G bytes

Figure 4.7. Byte Capacity of a File – 1K Bytes Per Block

Directoare Unix

- implementate ca perechi (nr inode, nume)
  - interpretare: director ca tabela de simboli
- contin automat intrari pt. "." si ".."
- *link*
  - intrari in directoare diferite care refera acelasi inode
  - *hard*: intrari pe acelasi sistem de fisiere (disc)
  - *soft*: intrari pe diferite sisteme de fisiere (discuri), de fapt fisiere speciale in care continutul fisierului este numele fisierului linkat
- *metadate*
  - blocuri folosite pt a stoca informatii administrative despre sistemul de fisiere si NU date obisnuite din fisiere
  - ex: superblock, free list-uri, inode, blocuri indirecte, etc

Virtual Filesystem Switch (VFS)

- inainte de a fi folosite, sistemele de fisiere se instaleza in ierarhia de directoare prin operatia *mount*
  - discul formatat cu un anumit sistem de fisiere se asociaza ("incaleca") cu un subdirector (numit *mount point*) din ierarhia de directoare

  Ex:

  mount –t ext4 /dev/sda1 /mnt

  mount –t nfs 192:168.0.45:/home /home

  mount –t iso9660 –o loop ubuntu.iso /dev/cdrom

- in sistemele de operare moderne, multiple tipuri de sisteme de fisiere pot fi instalate in aceeasi ierarhie de directoare prin intermediul VFS

Virtual Filesystem Switch (VFS)

- VFS defineste abstractia de *v-node*
  - la instalarea unui sistem de fisiere, kernelul aloca o structura VFS care inregistreaza in campul *v_op* pointeri catre functiile necesare implementarii apelurilor de sistem (*open*, *close*, *read*, *write*, etc) pt acel tip de sistem de fisiere
  - de fapt, un fisier deschis e reprezentat in kernel de un *v-node* (nu *inode*), eventual proaspat alocat, daca fisierul nu exista
  - restul kernelului invoca functiile din *v_op* prin functii generice care redirectioneaza executia catre operatiile specifice sistemului de fisiere respective, eg

    *VOP_OPEN(vnode *vp) { *(vp->v_op->vop_open)(vp, …) ; }*

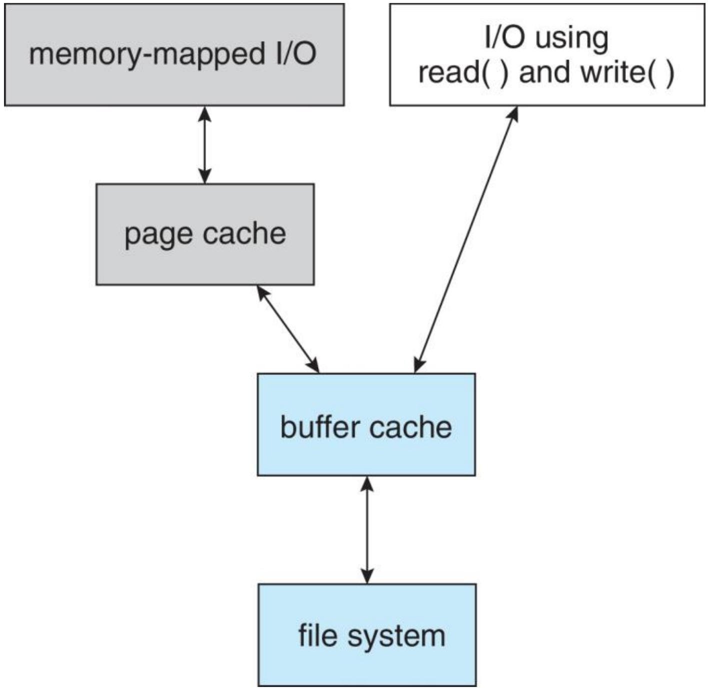
Performance

- Performance
  - Keeping data and metadata close together
  - **Buffer cache** – separate section of main memory for frequently used blocks
  - **Synchronous** writes sometimes requested by apps or needed by OS
    - No buffering / caching – writes must hit disk before acknowledgement
    - **Asynchronous** writes more common, buffer-able, faster
  - **Free-behind** and **read-ahead** – techniques to optimize sequential access
  - Reads frequently slower than writes

Page Cache

- A **page cache** caches pages rather than disk blocks using virtual memory techniques and addresses
- Memory-mapped I/O uses a page cache
- Routine I/O through the file system uses the buffer (disk) cache
- This leads to the following figure

I/O Without a Unified Buffer Cache



Unified Buffer Cache

- A **unified buffer cache** uses the same page cache to cache both memory-mapped pages and ordinary file system I/O to avoid **double caching**
- But which caches get priority, and what replacement algorithms to use?

I/O Using a Unified Buffer Cache